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Objectives

- 1 To explain the meaning and use of different concepts of cost.
- 2 To show how different concepts of cost are relevant for managerial decision-making.
- 3 To explain how production relationships underlie cost relationships.
- 4 To explain cost behaviour in the short run.
- 5 To explain cost behaviour in the long run.
- 6 To explain how cost relationships can be derived in mathematical terms.
- 7 To explain the purpose and principles of cost–volume–profit analysis.
- 8 To describe a problem-solving approach for applying cost–volume–profit analysis.

6.1 Introduction

6.1.1 Importance of costs for decision-making

Demand analysis is fundamentally concerned with the revenue side of an organization's operation; cost analysis is also vital in managerial economics, and managers must have a good understanding of cost relationships if they are to maximize the value of the firm. Many costs are more controllable than are factors affecting revenue. While a firm can estimate what effect an increase in advertising expenditure will have on sales revenue, this effect is generally

more uncertain than a decision to switch suppliers, or invest in new machinery, or close a plant.

Just as with production theory, the distinction between short run and long run is an important one. In the short run, managers are concerned with determining the optimal level of output to produce from a given plant size (or plant sizes, for a multiplant firm), and then planning production accordingly, in terms of the optimal input of the variable factor, scheduling and so on. In the long run, all inputs are variable so the most fundamental decision the firm has to make is the scale at which to operate. The optimal scale is the one that is the most efficient, in economic terms, for producing a given output.

Cost analysis is made complex because there are many different definitions and concepts of cost, and it is not always straightforward to determine which costs to use and how to measure them in a particular situation. The focus here is on the **relevant costs for decision-making**. In order to clarify this aspect the following four distinctions are important.

6.1.2 *Explicit and implicit costs*

Explicit costs can be considered as **expenses** or out-of-pocket costs (rent, raw materials, fuel, wages); they are normally recorded in a firm's accounts. However, the economic cost of using a resource is its **opportunity cost**, which is **the cost of forgoing the next most profitable use of the resource, or the benefit that could be obtained from the next-best use**. This involves both explicit and implicit costs. Let us take the example of a student considering undertaking an MBA; the relevant costs can be classified as either explicit costs or implicit costs.

Explicit costs include fees, books, accommodation, food, transportation, recreation and entertainment and so on. Not all of these may be directly related to doing an MBA, the last category for example, so they can be regarded as incidental costs. Money still has to be made available to pay these costs.

Implicit costs are non-cash costs, like the salary that could have been earned, leisure time forgone (if work required on the MBA exceeds the hours of salaried work), and interest forgone on assets which have to be used to pay MBA expenses.

Opportunity costs would include elements of both, but are not simply the sum of the two; for example, accommodation is not an opportunity cost if the student would be in the same accommodation whether they were doing the MBA or not. Opportunity costs should be used for decision-making purposes, meaning making the fundamental decision whether to do the MBA or not. These costs then have to be compared with the expected benefits, monetary and non-monetary, of undertaking an MBA programme. This does not mean that the other costs are unimportant; they are still relevant in cash planning.

6.1.3 *Historical and current costs*

Historical costs represent **actual cash outlay** and this is what accountants record and measure. This means measuring costs in historical terms, at the time they

were incurred. Although this is relevant for tax purposes it may not reflect the **current costs**.

Current costs refer to **the amount that would be paid for an item under present market conditions**. Often current costs exceed historical costs, particularly with inflation. In some situations, for example IT equipment, current costs tend to be below historical costs because of rapid improvements in technology. In this case the item being costed may no longer be available, and the appropriate cost is the **replacement cost**. This is the cost of duplicating the productive capability of the item using current technology. Replacement cost is the relevant cost for decision-making. The following example illustrates this principle.

Assume that Clearglass Conservatories is offered a contract to build a conservatory at a property, at a price of £60,000. The labour costs are £40,000 and the materials necessary to complete the job are already in inventory, valued at the historical cost of £15,000. If the job is accepted, Clearglass, as an ongoing concern, will have to replace the materials, but the price of these has risen, so the current cost is £22,000. If Clearglass uses historical cost to cost the job they will accept it, expecting to make a profit of £5,000. They should, however, reject it since they will really make a loss of £2,000. This can be seen more clearly if we consider what happens if they accept the job and then restore their inventory to the previous level. They will end up receiving £60,000 and paying out £62,000 (£40,000 for labour and £22,000 for materials), thus losing £2,000.

6.1.4 Sunk and incremental costs

Sunk costs are **costs that do not vary according to different decisions**. An example was given earlier in the case of the MBA student's accommodation; the accommodation cost was the same whether or not the student did the MBA. Often these costs refer to outlays that have already occurred at the time of decision-making, like the cost of market research conducted before deciding whether to launch a new product.

Incremental costs refer to **changes in costs caused by a particular decision**. Using the same example, if the student would have to pay £4,000 for yearly accommodation doing a salaried job and £6,000 for accommodation to do the MBA, the incremental cost associated with the decision to do the MBA would be £2,000 (assuming simplistically that there are no other costs or benefits related to the differences in accommodation). Incremental costs are the relevant costs for decision-making.

6.1.5 Private and social costs

Private costs refer to costs that accrue directly to the individuals performing a particular activity, in other words they are **internal costs**. For private firms these are the only costs that are relevant, unless there are ethical considerations (see Chapter 12).

Social costs also include **external** costs that are passed on to other parties, and are often difficult to value. For example, motorists cause pollution and

congestion which affect many other people (this is the economic justification for fuel duties, which are an attempt to internalize these externalities, as seen in Chapter 12). Therefore when a resource like oil or petrol is used there are both internal and external costs. The social costs are the sum of the two, meaning the total cost to society of using a resource (being careful not to double-count any duties). Social costs are relevant for public policy decision-making. In this situation the technique of **cost-benefit analysis** is often used. However, since we are largely concerned with managerial decision-making, social costs and cost-benefit analysis will not be examined here.

6.1.6 Relevant costs for decision-making

For private firms it has been shown above that it is the opportunity costs, the replacement costs and the incremental costs that are relevant. These concepts are all illustrated in the following case study.

Case study 6.1: Brewster Roofing

Mr Brewster operates a roofing company in London and has been asked by the local government authority of Merton to repair the roofs of several of their properties damaged in a recent storm. The job must be completed during the next four weeks (twenty working days), and Merton has offered £16,000 for the job. Mr Brewster has estimated that the job requires seventy-five work days, but he can only use his regular three workers for the job because it is a very busy period for the industry as a whole. Fortunately, Mr Brewster's son, Will, can take time off from his regular job (paying £80 per day) to help complete the work. Mr Brewster has estimated that, for his regular employees, the cost per work day is £150. This consists of a wage of £100 (which is only paid if the employee is working) and £50 in contributions to the government (which are paid annually regardless of how many days the employees work).

Brewster Roofing has all the equipment necessary for the job and has some of the materials available in inventory. The materials cost £5,000 originally, but these costs have since increased by an average of 5

per cent. Additional materials costing £3,000 are also required.

Mr Brewster has costed the job as follows:

Revenue	£16,000
Costs	
Labour	£9,000
Materials	£8,000
Total costs	£17,000
Profit	(£1,000)

On the basis of the above analysis Mr Brewster rejects the job.

Questions

- 1 Prepare a revised cost estimate for the job, taking into account opportunity costs, replacement costs and incremental costs. Assume that Mr Brewster considers the job from the viewpoint of a family business, including himself and his son together.
- 2 Advise Mr Brewster regarding whether he should accept the job, stating any assumptions involved in your analysis.

6.1.7 Summary of cost concepts

Several points emerge from the above discussion of costs:

- 1 There is not always a right and a wrong way to use cost concepts. The right costs for decision-making are not the right costs for estimating tax liability and vice versa.

- 2 Managers must be very careful in using cost information prepared by accountants, since it has been collected and categorized for different purposes.
- 3 The determination of costs is not always purely objective; this is particularly true of implicit costs, where a considerable degree of judgement is often required.

6.2 Short-run cost behaviour

Managers want to know the nature of the cost functions pertaining to their firm for the following reasons:

- 1 To make pricing decisions – this aspect is considered in Chapters 8 and 10.
- 2 To determine the appropriate levels of other factors in the marketing mix – this is considered in Chapters 9 and 10.
- 3 To forecast and plan for the costs and input levels associated with a given level of output.

6.2.1 Classification of costs

It was seen in Chapter 5 that the short run in economic terms is defined as the period during which at least one factor of production is fixed and others are variable. This leads to a further classification of costs, into fixed costs and variable costs.

Fixed costs are **related to the fixed factors and do not vary with output in the short run**. Examples are rent, insurance, interest payments, and depreciation (if estimated on a time basis). These costs may vary in the short run, for example if the interest rate rises, but not because of a change in output. Fixed costs have to be paid even if output is zero for any period, for example when there is a strike.

Variable costs are **related to the variable factors and vary directly with output**. This was assumed in some of the analysis in Chapter 3, for example in Table 3.4 relating price changes to profit. Examples of variable costs are raw materials, wages, depreciation related to the use of equipment, and some fuel costs.

In practice a clear distinction between fixed and variable costs is not always possible; some costs, like fuel above, may have fixed and variable elements. Other costs, like administrative salaries, may be fixed over a certain output range, but if output exceeds the range an increase in staff may be required, thus increasing costs.

6.2.2 Types of unit cost

Managers are often more interested in unit costs than in total costs, for the following reasons:

- 1 Unit costs provide a better measure of the performance of the firm in terms of efficiency. This means that comparisons on a time-series or cross-section basis are easier to make.
- 2 Unit costs can be compared directly with price, and conclusions drawn regarding profit. This is explained in section 6.5 on CVP analysis.
- 3 It is easier to understand cost relationships with output in terms of unit costs.

There are four main types of unit cost in the short run:

- 1 *Marginal cost (MC)*. This can be defined as **the additional cost of producing an additional unit of output**. Thus we can write:

$$MC = \text{change in total costs} / \text{change in output} = \Delta C / \Delta Q \quad (6.1)$$

In the limiting case where the change in output is infinitely small we can write:

$$MC = dC/dQ \quad (6.2)$$

- 2 *Average variable cost (AVC)*. This refers to the total variable costs divided by output; thus we can write:

$$AVC = VC/Q \quad (6.3)$$

- 3 *Average fixed cost (AFC)*. This refers to total fixed costs divided by output; we can write:

$$AFC = FC/Q \quad (6.4)$$

- 4 *Average total cost (ATC)*. This refers to total costs divided by output; we can write:

$$ATC = TC/Q \quad (6.5)$$

or

$$ATC = (FC + VC)/Q = FC/Q + VC/Q = AFC + AVC \quad (6.6)$$

All of the above types of unit cost can be derived from the production table in Table 5.3, provided that we know the prices of the inputs, labour and capital.

6.2.3 Derivation of cost functions from production functions

In the example relating to Viking Shoes in Chapter 5 it was assumed that the price of labour was £400 per worker per week and the price of machines was £500 per week. We can now compute the various costs and unit costs related to the inputs and outputs in Table 5.3; these values are shown in Table 6.1. Note that once again the marginal values, in this case MC, have to be plotted corresponding to the mid-values of outputs. This is also important in plotting the values on the graph in Figure 6.1. The final row of Table 5.3 is omitted since the use of eight workers with three machines is not technically efficient.

Table 6.1. Viking shoes: short-run cost functions

Output, Q (units)	Capital, K (machines)	Labour, L (workers)	Fixed cost, FC	Variable cost, VC	Total cost, TC	$AFC = FC/Q$	$AVC = VC/Q$	$ATC = TC/Q$	$MC = \Delta C/\Delta Q$
0	3	0	1,500	0	1,500	—	—	—	
13	3	1	1,500	400	1,900	115	31	146	31
27	3	2	1,500	800	2,300	56	30	85	29
39	3	3	1,500	1,200	2,700	38	31	69	33
50	3	4	1,500	1,600	3,100	30	32	62	36
59	3	5	1,500	2,000	3,500	25	34	59	44
64	3	6	1,500	2,400	3,900	23	37	61	80
66	3	7	1,500	2,800	4,300	23	42	65	200

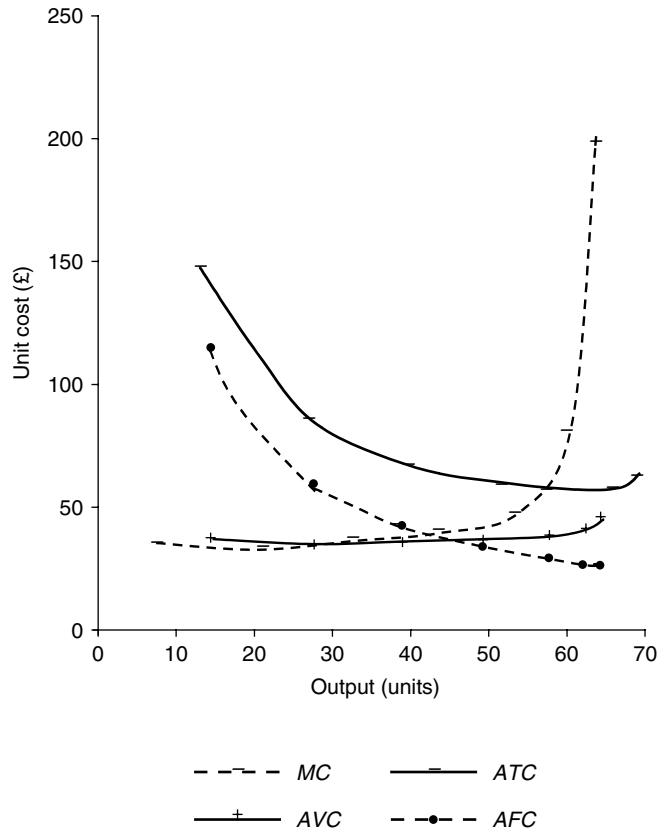


Figure 6.1. Viking shoes: unit cost functions.

The four unit cost functions in the last four columns of Table 6.1 can be illustrated in a graph. They are shown in Figure 6.1. The relationships between unit costs and output are now explained, with the aid of Figure 6.2. This shows the unit cost relationships in general terms, and their correspondence to production relationships and total cost relationships.

6.2.4 Factors determining relationships with output

It is helpful to begin the explanation of unit cost behaviour with marginal cost since this determines the shape of both the AVC and ATC curves.

a. Marginal cost

There are two features of the MC curve that need to be explained.

1. The MC curve is U-shaped. MC falls to begin with as output rises, because of increasing returns, and then, after reaching output Q_1 , it begins to rise because of diminishing returns. This is best explained by considering the following relationship:

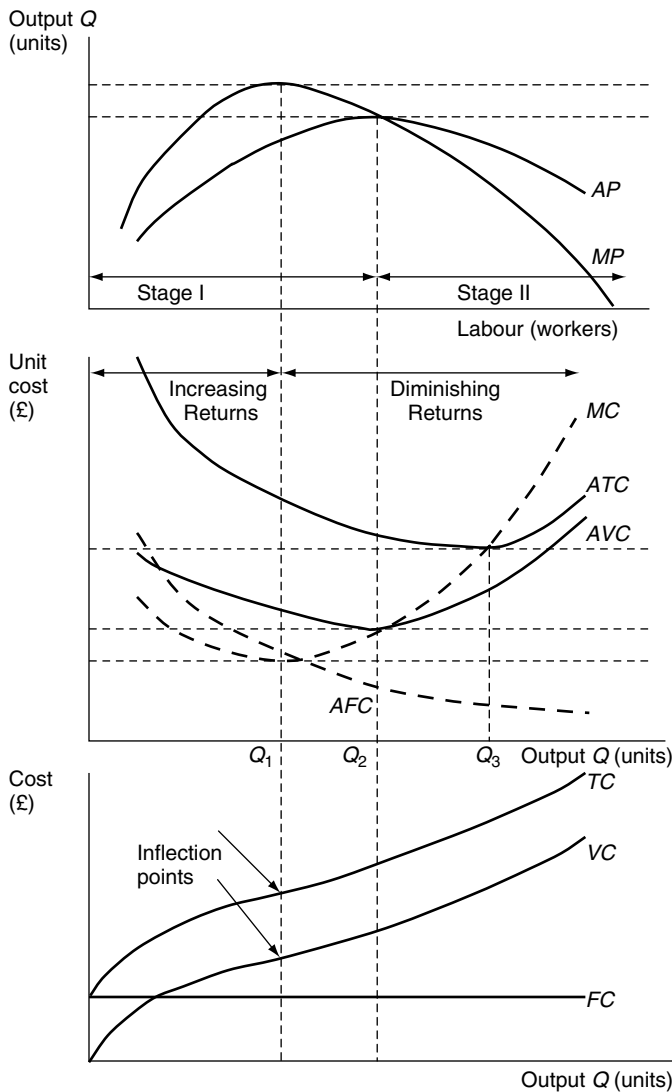


Figure 6.2. Short-run cost and production functions.

$$MC = \Delta C / \Delta Q = \frac{\Delta C / \Delta L}{\Delta Q / \Delta L}$$

where $\Delta C / \Delta L$ = the additional cost of hiring one more worker, in other words the price of labour, P_L , and $\Delta Q / \Delta L$ = the marginal product of labour MP_L .

Thus we can write:

$$MC = \frac{P_L}{MP_L} \quad (6.7)$$

Assuming that each additional worker is paid the same wage, P_L is constant. This means that MC is inversely related to marginal product. Thus, when there are increasing returns and MP_L is rising, it follows that MC must be falling; after output Q_1 there are diminishing returns and MP_L is falling, so MC must be rising.

2. The MC curve intersects both the AVC and ATC curves at their minimum points. The basic relationships between average and marginal values have already been explained in the previous chapter, in section 5.3. A similar relationship exists for marginal costs as for marginal product, in terms of its relationship with average costs. Thus when marginal costs are less than average costs, average costs must be falling and vice versa. Average fixed cost is unaffected by marginal cost, by definition.

b. Average variable cost

The AVC curve is also U-shaped; this is for similar reasons as for the MC curve. It can be shown that:

$$AVC = VC/Q = \frac{VC/L}{Q/L} = \frac{P_L}{AP_L} \quad (6.8)$$

Thus average variable cost is inversely proportional to average product; when AP is rising, AVC must be falling and vice versa. The range of output when AP is rising, and thus AVC falling (up to Q_2), is the range of operation of stage I. Once again increasing and diminishing returns are responsible for this behaviour.

c. Average fixed cost

As output rises the fixed costs become spread over a larger level of output, thus causing AFC to fall continuously throughout the output range. The concept of spreading fixed cost is a fundamental one in managerial economics and is relevant in many everyday activities as well. An example is a consumer buying a season ticket; it does not matter whether the ticket is for public transportation or going to see football matches, the principle is the same. Unless the consumer is going to use the ticket frequently it is not worth buying, since they will not be able to spread the fixed cost of the ticket over a sufficiently large output, meaning trips or matches.

d. Average total Cost

This is also U-shaped because it is the sum of AVC and AFC . These are summed vertically on the graph. At low levels of output, ATC falls because both AVC and AFC are falling. Beyond the level of output Q_2 , AVC starts to rise while AFC continues to fall. Beyond output Q_3 , AVC rises faster than AFC falls; this means that the effect of diminishing returns more than offsets the spreading of fixed costs, causing ATC to rise. The significance of output Q_3 in terms of efficiency is explained in the next subsection.

6.2.5 Efficiency

As was seen in the last chapter, there are various different concepts of efficiency in economics. Since it was assumed that all the output values in the production functions shown in Tables 5.1 and 5.3 involved technical efficiency, it follows that all the outputs in Figure 5.1 also involve technical efficiency since they are based on the same production function. It is also true that all these outputs involve economic efficiency, in that they all involve a least-cost combination of inputs in the short run, *assuming a given output*. However, only one of these outputs results in the unit cost (average total cost) being minimized if we no longer assume a given output, and that is output Q_3 . It can therefore be said that Q_3 is the most efficient level of output in the short run for a firm of given plant size(s). Unfortunately, in many microeconomics texts this output is referred to as the ‘maximum capacity’ rather than maximum efficiency; this terminology is confusing. It is more accurate to apply the term ‘maximum capacity’ to the maximum output for the firm in the short run. This does not imply that the most efficient output is an optimal output as far as the firm is concerned, because only costs are considered, not revenue; therefore it tells us nothing about profit.

This type of efficiency in the short run can be viewed as **a compromise between spreading fixed costs and getting diminishing returns**. If more output than Q_3 is produced, less is gained from the further spreading of fixed costs than is lost from getting diminishing returns; therefore *ATC* rises beyond this level of output.

6.2.6 Changes in input prices

Since it has been assumed that the firm is operating at technical and economic efficiency, it cannot reduce the controllable elements of its cost structure. However, some of its costs are uncontrollable and may rise or fall in the short run. For example, the interest rate on the firm’s borrowing may rise, and this will have an immediate effect on the unit cost curves. Since it has been assumed that capital is a fixed factor, the increased interest cost represents an increase in fixed cost. This will shift the *AFC* and *ATC* curves upwards, while *AVC* and *MC* remain unchanged.

Likewise, if the price of raw materials rises, this represents an increase in a variable cost and will therefore shift the *MC*, *AVC* and *ATC* curves upwards. A decrease in price will similarly shift the same curves downwards.

6.2.7 Different forms of cost function

In most economic analysis, the form of short-run cost function that is most frequently used is the cubic total cost function. This is the form that gives rise to the shapes of function shown in Figures 6.1 and 6.2.

a. Cubic cost functions

These are related to the cubic production functions examined in the previous chapter. The relationship between the two types of function can be seen most easily in the relationships between marginal product and marginal cost. Both of these are quadratic functions that are inversely related, as shown in (6.7), and this is illustrated graphically in Figure 6.2.

Two other forms of total cost function are also sometimes used in economic analysis; these are the **quadratic** form and the **linear** form. These forms and their implications in terms of unit costs and graphical shapes are now considered.

b. Quadratic cost functions*

These have the general form:

$$C = a + bQ + cQ^2 \quad (6.9)$$

In this case the marginal cost is given by:

$$MC = b + 2cQ \quad (6.10)$$

This is a linear function, as shown in Figure 6.3. The economic implication of this is that the law of diminishing returns takes effect as soon as the firm starts to produce.

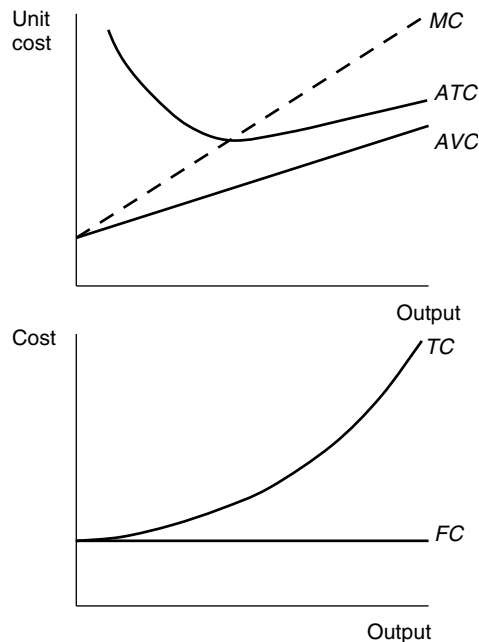


Figure 6.3. The quadratic cost function.

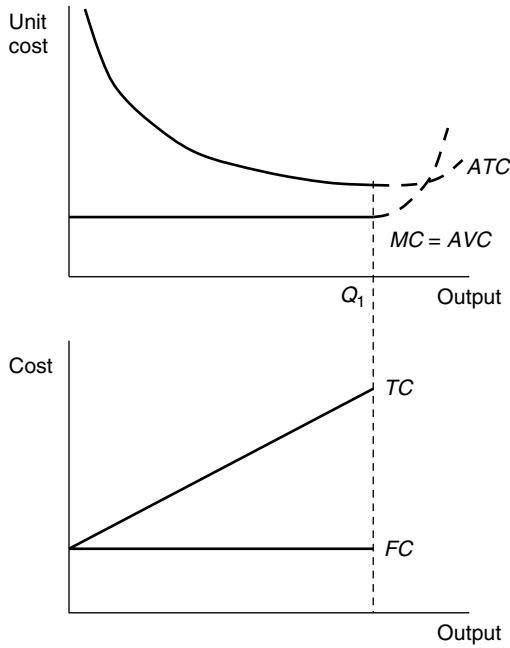


Figure 6.4. The linear cost function.

It can also be shown that a quadratic cost function will occur if the production function is of the Cobb–Douglas type with the output elasticity of labour equal to 0.5:

$$Q = aL^{0.5}K^c \quad (6.11)$$

From this an expression for L can be derived as follows:

$$L = (aK^c)^{-2}Q^2$$

Since $C = P_L L + P_K K$, we can now express C as the following quadratic function:

$$C = P_K K + P_L (aK^c)^{-2}Q^2 \quad (6.12)$$

In more general terms, with any Cobb–Douglas production function $Q = aL^b K^c$, the short-run cost function will be a polynomial of degree b^{-1} .

c. Linear cost functions

These have the general form:

$$C = a + bQ \quad (6.13)$$

In this case the marginal cost is given by:

$$MC = b \quad (6.14)$$

This is obviously the simplest form of cost function from a mathematical viewpoint, but it is also important empirically. As seen in more detail in

Chapter 7, many firms do observe this type of cost relationship with output in the short run; this is the reason why this form of function is most frequently used in cost-volume-profit analysis, the subject of section 6.5.

What are the economic implications of this type of function? Since marginal cost is constant it means that the firm does not encounter either increasing or diminishing returns, *at least over its normal output range*. It is this last condition that gives this mathematical form its empirical validity. As seen in Figure 6.4, the MC curve is horizontal, but this does not mean that the curve would remain horizontal over all outputs. The firm's normal output range may extend up to Q_1 ; beyond this level of output it is possible for diminishing returns to occur as the firm approaches full capacity. This possibility is illustrated by extending the unit cost curves using dashed lines. A further implication of this type of cost function is that AVC is also given by the constant b . Thus MC and AVC are both equal to each other and constant.

6.3 Long-run cost behaviour

In the long run the firm can change all its inputs; therefore all costs are variable. In this context the main decision that managers must make is the determination of the optimal plant size or scale of the firm, given a certain target level of output. Once the optimal combination of inputs is selected, some of these inputs, like plant and equipment, become fixed in the short run.

6.3.1 Derivation of cost functions from production functions*

This analysis has already been considered for short-run functions. In the long run the situation is more complicated mathematically, since all the inputs are variable. However, for the commonly considered cubic function in Table 5.1, it was seen that there were increasing returns to scale to begin with but that at higher levels of output there were decreasing returns to scale. In more general terms the type of cost function can be derived from the firm's expansion path, as shown in Figure 5.9; this shows various least-cost combinations of inputs for producing given outputs in the long run.

With the Cobb-Douglas production function it can be shown that the long-run cost function is a power function. More precisely, with any Cobb-Douglas production function $Q = aL^bK^c$, the mathematical form of the cost function is as follows:

$$C = a'Q^{1/(b+c)} \quad (6.15)$$

The proof of this derivation is beyond the scope of this text,¹ and involves Lagrangian multiplier analysis. However, the economic implications are important. It has already been seen that the powers in the Cobb-Douglas production function represent output elasticities, which, when summed

together, indicate the type of returns to scale. For example, if the output elasticities sum to one, there are constant returns to scale: a 1 per cent increase in all inputs will cause output to increase by 1 per cent. From (6.15) it can be seen that in this case the power of Q will also be equal to 1; this value represents the cost elasticity with respect to output and again is an indicator of returns to scale, assuming that input prices remain constant. If the output elasticities sum to more than 1 it follows that the cost elasticity will be less than 1; this is again an indicator of returns to scale, in this case increasing returns to scale. Likewise, if the output elasticities sum to less than 1, it follows that the cost elasticity is more than 1 and there are decreasing returns to scale. Unlike the cubic cost function the returns to scale are the same at all levels of output.

However, an assumption in all of the above analysis is that input prices remain constant. In the long run the cost elasticity will depend not only on physical returns to scale but on monetary aspects as well: unit costs may rise or fall depending on the scale of the firm independently of the physical returns to scale. Both of these aspects, physical and monetary, now need to be examined in detail.

6.3.2 Economies of scale

Economies of scale (EOS) can be defined as **aspects of increasing scale that lead to falling long-run unit costs**. They can be classified in various ways, as discussed in the following three paragraphs.

1. *Internal/external*. Internal economies **arise from the growth of the firm itself**; in this sense they are controllable and under the influence of management decision-making. External economies **arise from the growth of the industry**, and are independent of the size of the firm. They are therefore further removed from managerial decision-making, though not entirely so; location decisions in particular may depend on these economies. External economies are sometimes called **economies of concentration** because they tend to arise when firms in the same industry are located close together. Further discussion of external economies and diseconomies is given in Chapter 8. Internal economies of scale are usually more important in terms of their effects on unit costs. Therefore, in view of these factors and the aspects of management control, it is the internal economies that we will concentrate on here.

2. *Physical/monetary*. This distinction has already been mentioned above, but will become clearer with the examples explained shortly. Physical economies cause increasing returns to scale; monetary economies reduce input prices.

3. *Level: product, plant and firm*. Again the distinction between these different levels at which economies occur will become clearer with the examples below. In general some of the cost advantages arise from producing more of one product, some from producing with a larger plant size, and some from producing with a larger firm.

There are four main categories of internal economies of scale:²

a. Technical economies

These arise mainly from *increased specialization* and *indivisibilities*. Larger firms can make use of more specialized equipment and labour in the production process, for example by using assembly lines. Virtually every product that is produced for the mass market, from jeans to CDs, computer chips to bottled soft drinks, is produced on some kind of assembly line. This has the advantage of increasing both labour and capital productivity. Such processes need a large initial investment, because they cannot perform the relevant functions on a small scale; thus indivisibilities are involved. A good example of this is car production; assembly lines for producing cars have to be very large in order to perform all the necessary tasks. Low-volume car producers, like the British companies Morgan and TVR, cannot afford to use assembly-line methods because they would not be able to spread the high level of fixed cost. These economies are physical and occur at the level of the product.

Indivisibilities also occur in other forms: larger firms are able to use more expensive but often more effective advertising media, like television, and they can afford to undertake research and development activities that small firms could not afford. Such economies are at the level of the firm as a whole.

Other technical economies relate to *increased dimensions*: as size increases, volume increases more rapidly than surface area. Since volume often determines output while surface area determines cost, unit costs fall as size increases. This is particularly important in the transportation and storage industries, and explains the development of jumbo (now super-jumbo) jets and supertankers. Another example is the trucks used to transport rocks and other materials in mining; in the 1950s the average capacity of these vehicles in the United States was 20 tons, by the 1970s it was 150 tons, and it is now about 300 tons. These economies occur mainly at the level of the plant.

A final technical economy relates to *massed resources*: larger firms find it easier to combine equipment or facilities with different capacities more efficiently, with less idle capacity. For example, a car manufacturer needs to use an engine-block casting facility, a panel-pressing facility, a paint shop and various machining and assembly facilities; these all tend to be of different sizes for technical reasons. Similarly, larger firms have a proportionately lower need for reserves of spare parts and maintenance workers. These economies are sometimes called *multiplant* economies, as opposed to *intraplant* economies, because they occur at the level of the firm rather than the individual plant.

These technical economies tend to be the most important source of economies of scale for most firms. They occur particularly in mass manufacturing, public utilities and mass transportation.

b. Managerial economies

Large firms find it easier to attract and use more specialized managers, who are more skilled and productive at performing specific managerial functions. Thus

a small firm may employ a general manager for all managerial functions; a mid-sized firm may employ separate managers for the main managerial functions of production, marketing, finance and human resources; a large firm may employ various managers within the marketing department, for example in purchasing, advertising, sales, public relations and market research. Like the technical economies these are physical in nature; they also occur mainly at the level of the whole firm.

c. Marketing economies

These relate mainly to obtaining bulk discounts; by buying in bulk larger firms can often enable their suppliers to obtain the technical economies of scale above. These discounts relate not just to buying raw materials and components but also to buying advertising. For example, if a firm buys twice as much advertising space or time, the total cost will usually less than double, thus unit costs will fall (assuming the firm sells twice as much). This type of economy of scale is obviously of a monetary nature.

d. Financial economies

The most obvious factor here is that large firms can often borrow at a lower interest rate, because they have a better credit rating, representing a lower default premium. In addition they have more sources of finance; they can use the capital markets, for example by issuing commercial paper, bonds and shares. These forms of raising finance often involve a lower cost of capital. Finally, a larger firm can enable its supplier of funds, normally a bank, to gain economies of scale of an administrative nature. Again these economies are clearly monetary and at the level of the whole firm.

The above economies of scale all represent cost advantages that larger firms can gain. There may be additional advantages in terms of the learning curve, described in a later section, and in terms of gaining monopoly power in the market. This last advantage is a demand rather than a cost advantage, and is discussed in Chapter 8.

6.3.3 Diseconomies of scale

Diseconomies of scale (DOS) are *aspects of increasing scale that lead to rising long-run unit costs*. Again they can be internal or external, physical or monetary, and can arise at the level of product, plant or firm. There are again four main sources of diseconomies of scale, though these do not correspond exactly to the four categories of economies of scale described above.

a. Technical diseconomies

Increased specialization can lead to problems as well as benefits. Workers doing repetitive jobs can suffer from low motivation, which reduces productivity and increases the chance of industrial unrest. The number of days lost

through strikes tends to be higher in industries that feature such processes, for example car manufacturing, mining, engineering and transportation and communications.³ Furthermore, a stoppage in such industries, whether caused by industrial unrest or by some other event like a machine breakdown, can cause the whole production process to come to a halt because of the interdependence of operations.

b. Managerial diseconomies

Large firms are more difficult to manage because communications tend to break down, both vertically and horizontally. This creates inefficiencies as co-operation and co-ordination within the firm suffer. Firms may try to combat this tendency by employing more administrative workers, but this is also going to increase unit costs. This communications problem is a major reason why many large firms are trying to contract services out to other firms and create flatter organizational structures, for example IBM and GM.

c. Marketing diseconomies

Although larger firms can often gain discounts in buying raw materials in bulk, there may be offsetting disadvantages of buying inputs in large quantities. If the firm is relying on local sources that are in limited supply, the high demand may drive up the price of such inputs; for example, the firm may have to offer higher wages to attract the desired quantity of workers.

d. Transportation diseconomies

Larger firms, particularly if they only use one plant, may face additional transportation costs as they try to increase the size of their market; the average transportation distance of goods to customers will increase. Again the above diseconomies relate only to cost disadvantages of large firms, or conversely the cost advantages of small firms. Larger firms may have other disadvantages, in terms of having less flexibility, a slower speed of response to environmental changes and the reduced ability to offer personal service to their customers.

6.3.4 Economies of scope

Whereas economies of scale relate to cost reductions caused by increasing scale, **economies of scope** occur when **changing the mix of operations has cost benefits**. For example, producing 100,000 units of product X may involve a unit cost of £100 if X is produced by itself; but if 100,000 units of X are produced along with a quantity of product Y, then the unit cost of producing X may fall. The same may happen to the unit cost of Y compared with producing it by itself. There are two main causes of this:

- 1 The products may use common processing facilities; for example, different car models being produced at the same plant.
- 2 There may be cost complementarity, especially when there are joint products or by-products, for example petrochemicals.

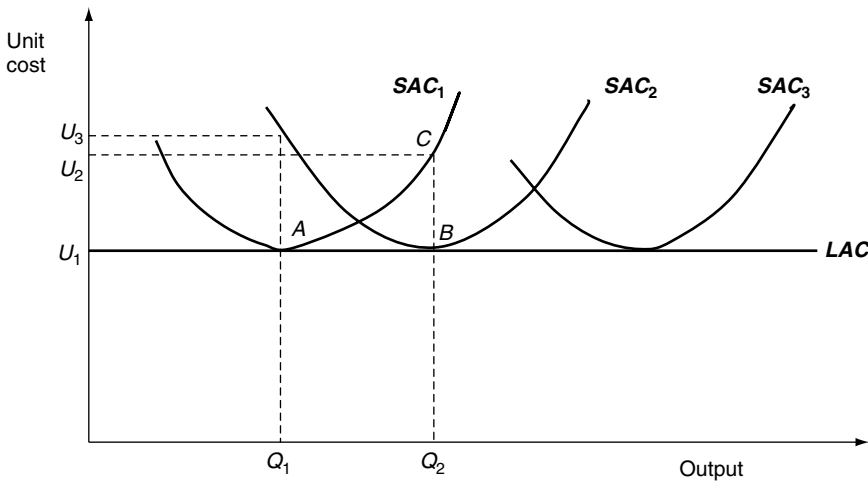


Figure 6.5. Short-run and long-run average cost functions in the absence of EOS and DOS.

It is also possible for a firm to experience diseconomies of scope if it sells products that ‘clash’ with each other in some way. An example would be EMI Records in the late 1970s when they signed the punk rock band the Sex Pistols, who openly advocated the use of violence. Since EMI was also marketing hospital equipment at the time, it faced considerable criticism and decided to drop the recording contract, at considerable cost.

The extent of economies of scope can be measured by estimating the percentage cost reduction caused by joint production, as follows:

$$S = \frac{C(Q_1) + C(Q_2) - C(Q_1 + Q_2)}{C(Q_1 + Q_2)} \quad (6.16)$$

where $C(Q_1)$ and $C(Q_2)$ represent the costs of producing outputs Q_1 and Q_2 independently, and $C(Q_1 + Q_2)$ represents the cost of producing outputs Q_1 and Q_2 jointly. If economies of scope exist, the joint cost is less than the sum of the individual costs, thus S is positive. The larger the value of S , the greater the economies of scope; if S is zero there are no economies of scope. Other measures of economies of scope are described in the next chapter.

6.3.5 Relationships between short- and long-run cost curves

In the long run the firm is able to use the least-cost combination of inputs to produce any given output, meaning that it can select the scale appropriate to its level of operation. This means that its long-run (LAC) curve is an envelope of its short-run average cost (SAC) curves. This is illustrated in Figures 6.5 and 6.6.

a. Optimal scale

In Figure 6.5 there are no economies or diseconomies of scale, thus unit costs are constant in the long run and the LAC curve is horizontal. Thus if the firm

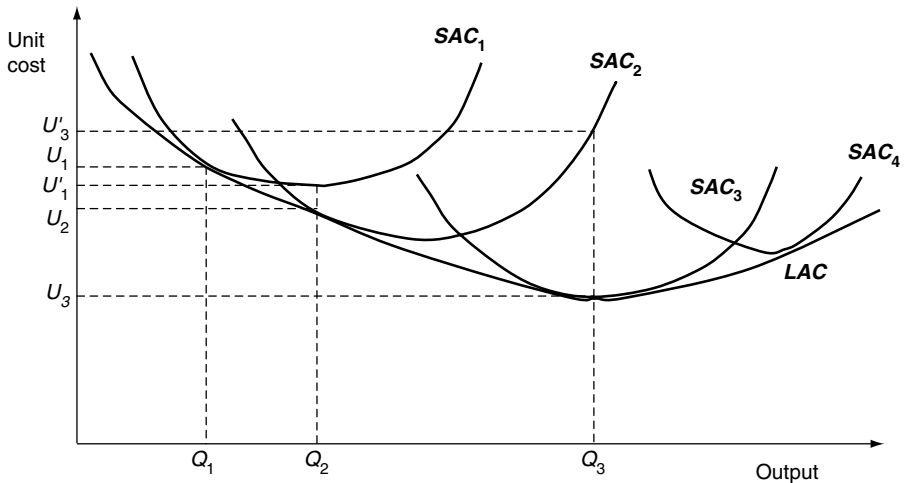


Figure 6.6. Short-run and long-run average cost functions with EOS and DOS.

wants to increase output from Q_1 to Q_2 in the long run it can move from point A to point B, and unit costs will remain constant at U_1 . On the other hand, if the firm wants to increase output from Q_1 to Q_2 in the short run it has to move from point A to point C on SAC_1 , with its unit costs rising to U_2 because it will be getting diminishing returns. We can say that for producing output Q_2 the scale or plant size associated with SAC_2 is the **optimal scale**; this means that this scale will result in the **minimum average cost**. However, for producing output Q_1 the scale associated with SAC_1 is preferable; the scale of SAC_2 is too large for this output, and the firm will be unable to spread its fixed costs sufficiently, unit costs being U_3 .

When there are economies or diseconomies of scale the situation is made more complicated. In Figure 6.6 there are economies of scale in the lower output range up to Q_3 , and diseconomies of scale in the output range above Q_3 . The optimal scale for producing output Q_1 is the scale associated with SAC_1 , since this is the scale that will produce Q_1 at the minimum unit cost (U_1). However, it should be noted that the reverse is not true: output Q_1 is not the optimal output for the scale of SAC_1 ; the optimal output for this scale is output Q_2 , since that is the most efficient output for this scale, at the minimum point of the SAC_1 curve, where unit costs are U'_1 .

The student may well ask at this point: what is going on here? Well, to be fair, even the economist Jacob Viner, who originally drew these graphs, did not know to begin with. The story goes that he asked his draughtsman to draw a long-run average cost curve as a U-shaped envelope going through all the minimum points of the short-run cost curves. The draughtsman then pointed out that this could not be done, but Viner insisted that the curve be drawn in this way. There was a sequel to this episode, which has become part of the legend relating to John Maynard Keynes. Viner also expressed his lack of admiration for Keynes's work in the 1930s, and when Keynes was asked

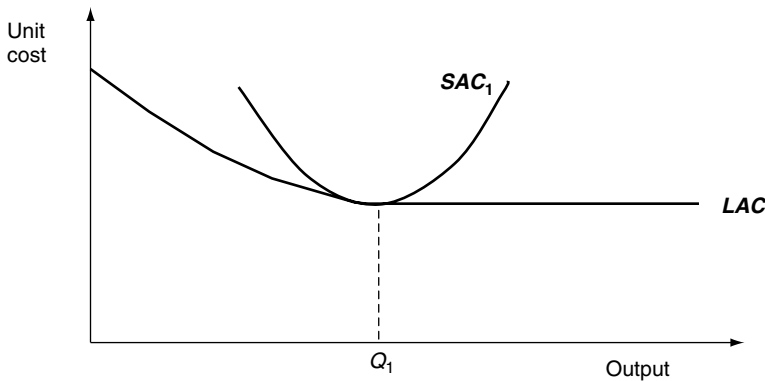


Figure 6.7. Minimum efficient scale.

whom he regarded as the greatest living economist, he replied that modesty forbade him from naming the greatest, but surely the second greatest was Jacob Viner's draughtsman!

It can be seen that only if the *LAC* curve is horizontal as in Figure 6.5 will it pass through all the minimum points of the *SAC* curves. If there are economies of scale it means that it is more efficient to use a larger scale at less than maximum efficiency than use a smaller scale at maximum efficiency, because more is gained from economies of scale than is lost from a failure to spread fixed cost. Thus in order to produce Q_2 it is better to use the scale of SAC_2 , resulting in unit cost U_2 , than to use the scale of SAC_1 , resulting in the higher unit cost U_1 .

We have to be careful therefore in using the concept of optimal scale or plant size; a scale can only be optimal for a given output. It is true that the most efficient scale in general is the one associated with SAC_3 , since this is the scale where unit costs are minimized in the long run. However, this scale would not be optimal for producing output Q_1 or Q_2 because it is too large and the firm would be unable to spread its fixed cost sufficiently. It may well happen that the size of the market is not sufficient to justify the use of a larger scale, even if such a scale is capable of reducing unit cost at a larger level of output.

b. Minimum efficient scale

The minimum efficient scale (MES) is defined as **the smallest scale at which long-run average cost is minimized**. In Figure 6.6 this is the scale of SAC_3 . However, as will be seen in the next chapter, not all *LAC* curves in practice turn out to be U-shaped; in some industries the *LAC* curve may take an L-shape; in this case the minimum efficient scale is where the curve flattens out, shown by output Q_1 in Figure 6.7.

c. Multiplant production

In the analysis to this point we have tended to use the terms 'scale' and 'plant size' interchangeably, although it has been pointed out that economies and

diseconomies of scale can occur at three different levels: product, plant and firm. For firms that operate just one plant the optimal plant size and optimal scale are synonymous. However, many firms operate more than one production facility. Sometimes these facilities involve the same stage in the production process, in which case the firm is said to be **horizontally integrated**. If the facilities operate at different stages of the production process, for example with a car manufacturer using different plants for pressing body panels, paint spraying and assembly, the firm is said to be **vertically integrated**. The analysis of optimal plant size is more complicated in this situation. It may be, for example, that the LAC curve is U-shaped if a firm operates a single plant, because as the plant size increases beyond a certain size, diseconomies at the plant level occur. The firm may be able to avoid these by using two or more smaller plants at minimum efficient scale; in effect this makes the LAC curve L-shaped and enables the firm to produce at lower unit cost.

6.3.6 Strategy implications

Managers must make decisions regarding optimal plant size and scale of operation in a dynamic environment. Demand may change unexpectedly, given the large range of uncontrollable factors affecting it discussed in Chapter 3. In this situation, managers must be prepared for the eventuality that there may be a mismatch between the quantity that the firm produces and the quantity that it sells.

If the LAC curve is U-shaped or L-shaped a firm might find that a change in demand might cause a problem in the short run, in that to meet the change in demand might require producing at above the minimum level of long-run average cost in the short run, in other words above the LAC curve. For example, referring to Figure 6.6, a firm may plan for output Q_2 and therefore operate at the scale of SAC_2 ; demand may then increase so that the quantity of Q_3 is demanded at the existing price. There are now various strategy options for the firm, each of which has certain advantages and disadvantages.

1. *Produce Q_3 with the existing scale.* Unit costs will rise in this case, from U_2 to U'_3 . The firm may therefore plan to increase its scale in the future, but only if it regards the increase in demand as being likely to continue.

2. *Continue to produce Q_2 , and use inventories to meet the excess demand.* In this case the firm is using inventories as a buffer to meet changes in demand, so that in the case of an unexpected shortfall in demand, inventory levels would be allowed to rise. This allows the firm some time to assess whether the change in demand is likely to be a lasting one. However, there is a cost: additional inventory levels involve additional holding cost.

3. *Continue to produce Q_2 , and use order backlogs to meet the excess demand.* This again is only appropriate if the increase in demand is seen as temporary. Otherwise the firm's order books will continue to increase in length, as will the waiting time for

customers to receive the product. The cost of this strategy is that increased customer waiting time will result in a loss of customers, so the firm will have to assess how sensitive its customers are to increases in waiting time.

4. *Increase the price to a level where only Q_2 will be demanded.* This may be regarded as a temporary measure if the increase in demand is seen as temporary; it may also be a temporary measure if the increase in demand is seen as a lasting one, while the firm takes the necessary time to increase its scale. The cost in this case is that the increase in price, even if temporary, may result in a loss of customers, who may not return later even if the price is reduced to its former level. The firm may lose some goodwill if it is seen as exploiting customers in the event of a shortage.

A more detailed discussion of the implications of the different strategies would require more information and the analysis introduced in Chapter 8.

6.4 The learning curve

There are many situations, not just in business but also in everyday life, where we learn better ways of doing something over time. Examples are playing tennis, using a keyboard, driving a car, or solving problems in managerial economics. In the workplace the factors that are involved are increased familiarization with the tasks involved, improvements in production methods, more efficient use of raw materials and machinery and fewer costly mistakes. These factors are obviously interdependent. The improvement in performance can be measured in a number of ways, depending on the situation, but the most common are in terms of unit cost or unit time to produce a product. This improvement is a function of experience, although when the concept was originally used in the 1930s in the analysis of aircraft production, the reason proposed for the learning curve effect was the learning by production workers, resulting in direct labour costs being reduced. It was only later that additional benefits in terms of production methods and indirect labour and other costs were considered. Hence, although the term **experience curve** is often used interchangeably with learning curve, some economists claim that there is some difference between the two concepts.⁴ Experience is normally measured in terms of cumulative output, that is the total output since production of a product began. Thus we can express the learning curve relationship as:

$$U = f(T) \quad (6.17)$$

where U is some measure of unit cost and T represents cumulative output.

It is important to note that this function is fundamentally different from the normal cost functions analysed so far, because of the nature of cumulative output. This means that the learning curve is neither a short-run nor a long-run cost function but can apply to any time horizon. Its effect is to cause the cost functions considered so far to shift downwards over time.

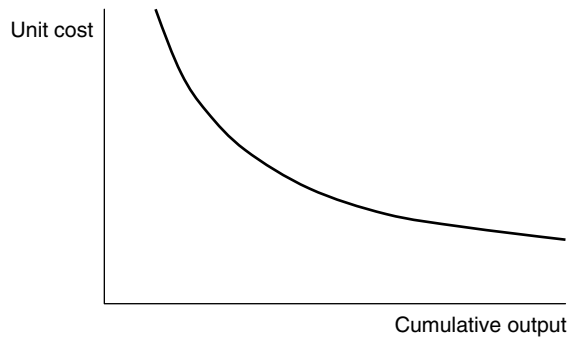


Figure 6.8. The learning curve.

We expect the relationship to be an inverse one if we learn from experience, but everyday experience of the experience curve(!) tells us that we tend to learn relatively quickly to begin with, and then our rate of learning slows down because of the familiar law of diminishing returns. Thus we would normally expect the situation that is shown in Figure 6.8.

This indicates that the appropriate mathematical form of the relationship is a power function:

$$U = aT^b \quad (6.18)$$

where U is normally measured as the marginal cost of a unit. The parameter b has a negative value, indicating an inverse relationship between marginal cost and cumulative output.

Learning curves are often described in terms of how a doubling of cumulative output affects marginal cost; if such a doubling results in a 20 per cent reduction in marginal cost it is said that the **learning rate** is 80 per cent. This learning rate is given by 2^b . The mathematical proof of this is as follows: let the initial cumulative output = T_1 and the initial marginal cost = U_1 ; then the marginal cost of double the output = $a(2T_1)^b = 2^b(aT_1^b) = 2^b(U_1)$.

This means that **every time cumulative output doubles the marginal cost becomes 2^b times what it was previously**. Other aspects of interpretation of the learning curve, its estimation and use in forecasting, and the results of empirical studies, are discussed in the next chapter, along with a numerical example.

6.5 Cost-volume-profit analysis

6.5.1 Purpose and assumptions

Cost-volume-profit (CVP) analysis examines relationships between costs, revenues and profit on the one hand and volume of output on the other. It is applied mainly to short-run situations. It is also sometimes referred to as

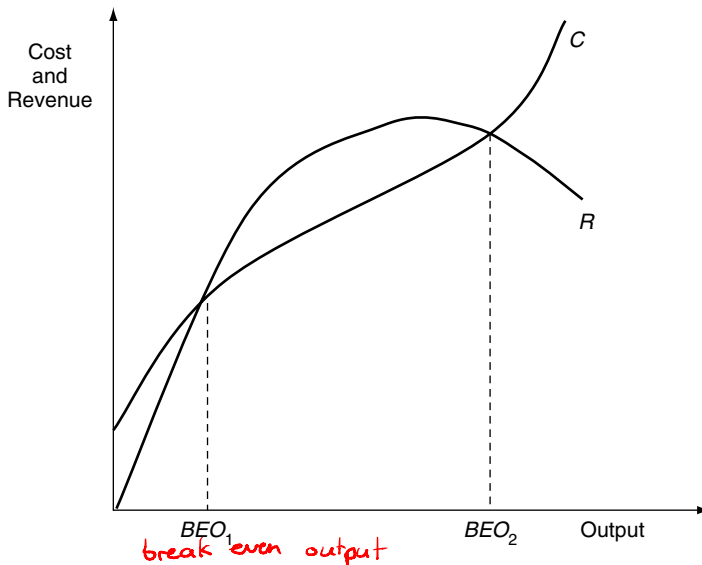


Figure 6.9. CVP graph: non-linear cost and revenue functions.

break-even analysis, but this term is somewhat narrow in scope; a firm's break-even output is specifically **the output where the firm makes zero profit**. CVP analysis covers a broader range of situations. The easiest approach is graphical.

On the basis of the analysis in the previous sections and assuming normal demand conditions (a downward-sloping demand curve), the graph shown in Figure 6.9 may result. In this situation the total cost curve is a cubic and the revenue curve is a quadratic. There will be two break-even outputs, and between them there will be a profit-maximizing output (not shown yet).

Although such situations can be analysed using CVP methods, it is more common to make the following assumptions:

- 1 MC is constant at all levels of output. This implies that the total cost function is linear, and the justification of this was given in subsection 6.2.7.
- 2 Firms are price-takers, meaning that they are operating under conditions of perfect competition.

The advantage of the above assumptions is that they result in linear total cost and revenue functions, thus simplifying the analysis mathematically. This situation is shown in Figure 6.10.

In this case there is only one break-even output; if the firm produces less than this it makes a loss (L_2 at output Q_2) and if it produces more it will make a profit (Π_1 at output Q_1). The more the firm produces the more profit it makes under the above assumptions.

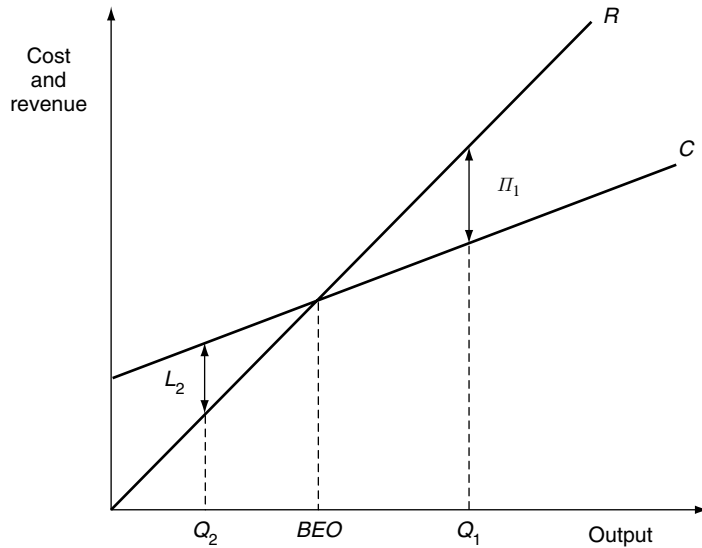


Figure 6.10. CVP graph: linear cost and revenue functions.

6.5.2 Break-even output

This can be derived mathematically, using linear cost and revenue functions. The revenue function, $R = PQ$, is set equal to the cost function, $C = a + bQ$, and we then solve for Q :

$$PQ = a + bQ$$

$$Q(P - b) = a$$

This gives the following expression for the break-even output:

$$BEO = a / (P - b) \quad (6.19)$$

where a = fixed cost, b = average variable cost, P = price. The denominator in this expression, $(P - b)$, is often referred to as the profit contribution.

6.5.3 Profit contribution

This is defined as **the money amount that each unit sold contributes to fixed costs and profit**. It is best explained with an example. A firm may have fixed costs of £10,000 per month and variable costs of £12 per unit; the market price may be £20.

$$\text{Profit contribution}(\Pi_C) = 20 - 12 = \text{£}8$$

To interpret this it is necessary to find the break-even output:

$$BEO = 10,000 / 8 = 1,250 \text{ units per month.}$$

If the firm produces 1,251 units, $\Pi = £8$; if it produces 1,252 units, $\Pi = £16$. Thus the firm is **contributing £8 to profit for every unit it produces above break-even output**.

If the firm produces no output it makes a loss of £10,000 (its fixed costs); if it produces 1 unit it makes a loss of £9,992. Thus the firm is **contributing £8 to fixed costs for every unit it produces below break-even output**.

6.5.4 Operating leverage*

The **degree of operating leverage** (DOL) refers to the **percentage change in profit resulting from a 1 per cent change in units sold**. This can be expressed mathematically as:

$$DOL = \frac{\% \Delta \Pi}{\% \Delta Q} \quad (6.20)$$

Thus it can be interpreted as an **elasticity of profits with respect to output** and can be measured in a similar way to the demand elasticities in Chapter 3:

$$DOL = \frac{d\Pi/\Pi}{dQ/Q} = d\Pi/dQ \times Q/\Pi \quad (6.21)$$

With linear cost and revenue functions the profit function must also be linear, therefore the *DOL* will vary with output. It will always be largest close to the break-even output. In the above example, if the firm is producing 1,500 units per month, the *DOL* is calculated as follows:

$$C = 10,000 + 12Q$$

$$R = 20Q$$

$$\Pi = 8Q - 10,000 = £2,000 \text{ at output of 1,500 units}$$

$$DOL = 8 \times 1,500/2,000 = 6$$

The interpretation of this result is that **a 1 per cent increase in output will increase profit by 6 per cent**.

6.5.5 Limitations of CVP analysis

CVP analysis can be a very useful aid to managerial decision-making, as illustrated in the following solved problems and case studies. There are, however, some important limitations of the technique. The main one is the set of restrictive assumptions on which it is often based. Profit does not usually increase linearly with output; many firms will have to reduce price in order to increase sales because they are not price-takers. Furthermore, as output increases they are likely to face diminishing returns as they approach capacity in the short term; inefficiencies and the payment of overtime wages may increase unit variable costs.

In the long run a number of factors may invalidate the simplified analysis above. The firm may change its capacity, thereby changing fixed costs. It may also change the quality of its products and its product mix. Therefore, as with other decision tools, CVP analysis must be used with care.

6.6 A problem-solving approach

This approach is designed as an aid to solving CVP problems. As with problems in demand theory students may ‘know’ the principles but sometimes do not know how to apply them to practical problems. Assuming linear cost and revenue functions, all CVP analysis is based on the following five equations:

1 Cost

$$C = a + bQ$$

(6.22)

2 Revenue

$$R = PQ$$

(6.23)

3 Profit

$$\Pi = R - C$$

(6.24)

4 Break-even

$$BEO = a / (P - b)$$

(6.25)

5 Profit contribution

$$\Pi_C = P - b$$

(6.26)

At this point it is necessary to review some basic algebraic principles. The equations often have to be **used in a sequence**. The key to success is using the equations in the **correct sequence**, starting off with values that are known and working towards the required unknown. If the equations are used in the wrong sequence there will be too many unknowns to solve the equation. However, in more complex problems, like SP6.2 below and the Nissan study (Case study 6.3), the existence of an equation with two or more unknowns does not necessarily create an impasse; it may be necessary to treat the problem as a **system of equations that need to be solved simultaneously**. In the case where there is an equation with two unknowns we have to find another equation in the same unknowns. If we have two equations in four unknowns then we need another two equations in the same unknowns to solve the system. The mathematical rule is that **we always need the same number of equations as unknowns in order to solve for the unknowns**. This does not guarantee a solution in general terms; it is a necessary condition, but not a sufficient one. However, in linear CVP problems the rule is a useful one.

There is one other useful aid to solving these more complex problems: it is often helpful to tabulate the data in a T-table with two columns, one relating to each entity, before processing it, as illustrated in SP6.2. This step can help to identify which equation or equations to use in order to begin the analysis. These principles are now applied in the following two problems.

SP6.1 CVP analysis

Last month Susie Q sold 24,000 litres of ice-cream. The variable costs were £2.70 per litre and each litre contributed 25 per cent of its revenue to fixed costs and profits. It has just discovered a new supplier which will enable it both to reduce its cost by £0.40 per litre and to improve its quality. However, it estimates that it will have to spend another £3,000 on advertising per month to inform customers of the improvement. Profits last month were £10,000.

- What is the previous month's cost function?
- What is the new cost function with the new supplier?
- How many litres will Susie Q have to sell to increase profit by 20 per cent, assuming it keeps its price the same?
- If Susie Q can raise its price by 10 per cent, what difference will this make to the sales in (c) above?

Solution

- a. $Q = 24,000$, $AVC = £2.70$, $\Pi_C = 0.25R/Q = 0.25P$, $\Pi = £10,000$

We want $C = a + bQ$.

We know $b = 2.7$, so have to find a .

Cannot use equation (4) because BEQ also unknown.

So we have to find C from (3), but this means finding R first from (2); in turn this means finding P from (5).

Thus the correct sequence of using the equations is (5), (2), (3), (1).

$$(5) \Pi_C = P - b$$

$$0.25P = P - 2.7$$

$$0.75P = 2.7$$

$$P = 3.6$$

$$(2) R = PQ$$

$$R = 3.6 \times 24,000 = 86,400$$

$$(3) \Pi = R - C$$

$$10,000 = 86,400 - C$$

$$C = 76,400$$

$$(1) 76,400 = a + 2.7 \times 24,000$$

$$76,400 = a + 64,800$$

$$a = 11,600$$

$$C = 11,600 + 2.7Q$$

- b. $C = 14,600 + 2.3Q$

- c. From (3), $12,000 = 3.6Q - (14,600 + 2.3Q)$

$$26,600 = 1.3Q$$

$$Q = 20,462 \text{ litres}$$

$$\begin{aligned} \text{d. } 12,000 &= 3.96Q - (14,600 + 2.3Q) \\ 26,600 &= 1.66Q \\ \mathbf{Q} &= \mathbf{16,024 \text{ litres}} \end{aligned}$$

SP6.2 CVP analysis

Lite and Brite produce lamps which they sell for £40. Lite has fixed costs of £8,000 less than Brite and average variable costs of £33, which is 10 per cent more than Brite. Lite has a break-even output which is 15 per cent less than Brite, and produces 25 per cent less revenue than Brite. Lite also makes £12,000 less profit than Brite.

- Calculate the cost functions of both firms.
- Calculate the outputs of both firms.
- Calculate the profits of both firms.

Solution

This type of problem, which we can call a ‘two-firm’ problem, is sometimes approached more easily by setting out the data in tabular form, as shown in Table 6.2; this helps us to see the relevant relationships more easily, thereby providing a handle to start the analysis. No calculations are performed at this stage.

$$\begin{aligned} \text{a.} \quad 33 &= 1.1(AVC_B) \\ AVC_B &= 30 \\ BEO_L &= FC_L / (40 - 33) \quad (1) \quad 7BEO_L = FC_L \\ BEO_B &= FC_B / (40 - 30) \quad (2) \quad 10BEO_B = FC_B \end{aligned}$$

We have two equations in four unknowns, so we need two more equations in the same unknowns:

$$(3) \quad FC_L = FC_B - 8,000$$

Table 6.2. CVP data For two-firm problem

	Lite	Brite
Price	£40	£40
FC	–£8,000	
AVC	£33	
AVC	+10 %	
BEO	–15 %	
R and Q	–25 %	
Profit		+£ 12,000

$$(4) \quad BEO_L = 0.85BEO_B$$

We need to reduce these four equations to one equation in one unknown, e.g. BEO_B .

$$7BEO_L = FC_B - 8,000$$

$$7(0.85BEO_B) = FC_B - 8,000$$

$$5.95BEO_B = 10BEO_B - 8,000$$

$$4.05BEO_B = 8,000$$

$$BEO_B = 1,975$$

$$FC_B = 19,753$$

$$C_L = 11,753 + 33Q_L \quad C_B = 19,753 + 30Q_B$$

$$b. (1) \quad \Pi_L = 7Q_L - 11,753 \quad (2) \quad \Pi_B = 10Q_B - 19,753$$

Again we have two equations in four unknowns, so we need two more in the same unknowns:

$$(3) \quad \Pi_L = \Pi_B - 12,000 \quad R_L = 0.75R_B, \text{ but } P_L = P_B$$

$$(4) \quad Q_L = 0.75 Q_B$$

$$7Q_L - 11,753 = 10Q_B - 19,753 - 12,000$$

$$20,000 = 10Q_B - 7Q_L = 10Q_B - 7(0.75Q_B) = 10Q_B - 5.25Q_B = 4.75Q_B$$

$$Q_B = 4,211$$

$$Q_L = 3,158$$

$$c. \quad \Pi_L = £10,353 \quad \Pi_B = £22,353$$

SP6.3 Mathematical analysis of cost relationships*

XTC Ltd has total costs of £45,000 and it is currently producing 5,000 units. It has examined its cost structure and has found that, of its variable costs, half vary in a linear relationship with output; the other variable costs increase by £1 for every 1,000-unit increase in output. Fixed costs are £10,000, and these determine the capacity of 6,000 units. The market price is £10.

- Determine the MC, AVC and ATC functions for the firm.
- Determine the current profit of the company.
- Determine the degree of operating leverage at current output.

Solution

a. $TVC = 45,000 - 10,000 = 35,000$

VC_1 has a linear relationship with output, so $AVC_1 = MC_1 = \text{constant}$.

$$MC_1 = 17,500/5,000 = 3.5$$

$MC_2 = a + 0.001Q$ where a is a constant

$$VC_2 = 17,500 \int (a + 0.001Q) = 17,500$$

$$[aQ + 0.0005Q^2] = 17,500$$

$$5,000a + 0.0005(5,000)^2 = 17,500$$

$$5,000a = 5,000$$

$$a = 1$$

$$MC_2 = 1 + 0.001Q$$

$$\mathbf{MC = 4.5 + 0.001Q}$$

$$TC = 10,000 + 4.5Q + 0.0005Q^2$$

$$\mathbf{AVC = 4.5 + 0.0005Q}$$

$$\mathbf{ATC = 10,000/Q + 4.5 + 0.0005Q}$$

b. $TC = 10,000 + 4.5(5,000) + 0.0005(5,000)^2$

$$TC = 45,000; TR = 10 \times 5,000 = 50,000$$

$$\mathbf{\Pi = £5,000}$$

c. $DOL = d\Pi/dQ \times Q/\Pi$

$$\Pi = 10Q - (10,000 + 4.5Q + 0.0005Q^2)$$

$$d\Pi/dQ = 5.5 - 0.001Q = 5.5 - 0.001(5,000) = 0.5$$

$$DOL = 0.5 \times 5,000/5,000$$

$$\mathbf{DOL = 0.5}$$

Case study 6.2: Converting to LPG – is it worth it?

Green fuel runs out of gas⁵

The cost of converting a car to run on liquefied petroleum gas (LPG) is about £1,500 in the UK, towards which a government grant would contribute about £700. From September 1 2004, LPG will on average cost 40.7p per litre, compared with 79.1p for ordinary unleaded petrol. However, LPG cars usually have slightly worse fuel consumption, losing about 13% in terms of miles per gallon.

Questions

Fast-Trak company owns a fleet of 20 cars, which are bought new and are used for 30,000 miles over two years before being sold off. The cars average 30 miles

to the gallon (imperial) on petrol. The conversion to LPG does not affect the price in the secondhand market.

- 1 Calculate the profit contribution per hundred miles of LPG compared with unleaded petrol, for one of Fast-Trak's cars.
- 2 Calculate the break-even mileage for the cars with the LPG conversion.
- 3 Calculate the effect on the profit of Fast-Trak of converting to LPG.
- 4 The government wants to encourage the use of LPG to protect the environment by reducing the break-even mileage to 10,000 miles; how large a grant should it offer for the LPG conversion?

Case study 6.3: Rescuing Nissan

On the rocky road to marriage⁶

A year ago it looked like mission impossible: rescuing Nissan after Renault had bought a controlling stake. Now Carlos Ghosn's recovery plan may be working, putting a full merger on the cards

Since last July, Mr Ghosn's job has been to rescue Japan's ailing car manufacturer; he is chief operating officer, but collects the title of president after this summer's general meeting. With the title goes a heavy responsibility. Nissan's plight is awful: losses in seven out of eight years, a domestic market share that drifted to an all-time low of 19% last year, and a pile of debt, totalling ¥1.4 trillion, plus ¥1.2 trillion in its financing division. 'It looked like mission impossible,' he admits, but we are on track to make profits in fiscal year 2000, ending next March. If we don't do that, we won't be credible.' [In 1999, Nissan's revenue was ¥6.3 trillion, and it made a loss of ¥30 billion.]

You do not have to be a rocket scientist to realise that a car maker able to make 2.4m vehicles a year in Japan, but producing fewer than 1.3m, is running its plants at a disastrously low level. To reach 75% utilisation, at which point most car firms break even, capacity needs to shrink by at least 30%. Under Mr Ghosn's plan, Nissan's domestic capacity is to

drop to a more realistic 1.65m units a year by 2002, maybe raising utilisation to 82%. First to go, by March 2001, will be the assembly plants at Murayama near Tokyo, the Nissan Shatai factory in Kyoto and the Aichi Kikai Minato plant in Nagoya. Next will be the Kurihama engine plant in Kanagawa and the Kyushu engine shop in Fukuoka.

Buttressed by his Renault experience, Mr Ghosn is likewise ignoring internal opposition to his changes at Nissan. His view is that the situation had become so desperate under the old regime that it had lost all credibility, so he has carte blanche to sort things out. But he is no autocrat. When he arrived last July, he formed nine cross-functional teams (CFTs) of middle managers to come up with plans to transform the company. According to Kiyoaki Sawada, a senior finance manager leading one team, these are not like ordinary project teams. 'We had those before, but everybody just represented their department's interest,' he says. The new teams are different.

The CFTs were a device that Mr Ghosn first used in America to bring about the merger of Michelin and another tyre company, Uniroyal Goodrich. He also installed them in Renault four years ago. He is hooked on the cross-functional approach, for several reasons. It works in a crisis, he says, because people

can understand the need for rapid action. It makes people act outside their specific areas. 'In most companies people make a specific contribution to the company in their function,' he says. 'But it is not expressed in terms of profit, only in terms of performing their function better.' Instead Mr Ghosn, who meets all the teams monthly, gets them to focus on profit creation, which he reckons lies in the interstices of different company functions. 'Profit is the most global aspect of a business, and it is cross-functional.'

The first product of the teams helped to form the basis of the Nissan revival plan unveiled last October. In the land of lifetime employment (at least for many workers in big companies), Mr Ghosn shocked Japan by announcing the closure of five factories employing over 16,000 people in Japan alone, cutting capacity by 30% to bring it more into line with sales and boosting utilisation rates to around the 80% rate.

Already machinery is being moved out of the doomed plants into those that will survive, and some workers have been transferred. Strong demand in the American market (and to a lesser extent in Europe) means that production is actually running about 10% higher than last year, so extra labour is needed in some other Japanese factories. Mr Ghosn hopes to avoid actually sacking workers, which is expensive in Japan. The second phase of his plan – to rationalise the Nissan and Renault distribution and dealer networks in Europe – has just been announced, and

aims to produce savings of about \$1 billion. Mr Ghosn hopes to cut costs at Nissan's British plant, already Europe's most efficient, by 30%, but thinks more pain is inevitable.

Questions

Assume:

- The second phase of the Ghosn plan does not come into effect until after the year 2000.
- The target operating profit for 2000 is 100 billion yen.
- Nissan earns 70 per cent of its revenues from vehicle sales and that other operations break even and will do the same in 2000.
- Average prices of vehicles sold are kept at the same level as 1999.
- Taxes are not included in the figures given.

- Calculate the average price of vehicles sold.
- Calculate average variable costs for 1999 and the target for 2000.
- Calculate the size of the overall Japanese vehicle market in 1999.
- If Nissan can reduce its variable costs in vehicle production by 5 per cent in 2000 compared with its target, estimate the effect on profit and return on sales.

Explain any assumptions in the above analysis.

Case study 6.4: Earls Court Gym

The management at Easyloan Bank is interested in offering a corporate fitness programme for its employees. It has decided to offer a 50 per cent subsidy to all who participate. The gym has agreed to offer the bank a 20 per cent discount on all fees for its employees, provided that they all have the same terms of membership. The bank's management observe from the membership information (Table 6.3) that there are various membership options. From a survey of its employees it establishes that they would use the

gym only. Of their visits, 75 per cent would be during weekdays and 25 per cent at weekends. Half their weekday visits would be before 12 noon and the other half would be after 5 p.m.

Questions

- Identify the feasible membership options and express these as cost functions for the bank.
- Draw a graph showing the above options and explain how the bank's management should select the optimal one.

Table 6.3. Earls Court Gym: membership information

Earls Court Gym P.O. Box 755 254 Earls Court Road London SW5 9AD Tel: 020-7370 1402					
Membership Rates					
Membership type	Period	Unrestricted access (i.e. anytime during opening hours)		Restricted access (i.e. Mon. to Fri. 9 a.m. to 5 p.m. & anytime at weekends)	
		New member	Renewal	New member	Renewal
Gym & Aerobics <i>Full use of gym and all exercise classes, plus shower, sauna and changing facilities</i>	12 months	£450	£416	£292	£270
	9 months	£399	£368	£261	£241
	6 months	£283	£260	£182	£167
	5 months	£268		£172	
	4 months	£243		£158	
	3 months	£199		£129	
	2 months	£153		£99	
	1 month	£85		£54	
	3 weeks	£66		£44	
	2 weeks	£46		£30	
	1 week	£27		£20	
	1 day	£10	£7 (Sat. & Sun.: £10)		
Gym only					
Unlimited use of gym and shower, sauna and changing facilities	12 months	£362	£335	£232	£216
	9 months	£318	£294	£210	£195
	6 months	£227	£211	£145	£134
	5 months	£214		£139	
	4 months	£195		£127	
	3 months	£160		£103	
	2 months	£123		£81	
	1 month	£67		£44	
	3 weeks	£53		£35	
	2 weeks	£38		£24	
	1 week	£21		£16	
	1 day	£3.50 (9 a.m. to 12 noon Mon. to Fri.) £5.00 (12 noon to 5 p.m. Mon. to Fri.) £7.50 (After 5 p.m. Mon. to Fri. and anytime Sat. and Sun.)			

Table 6.3. (cont.)

Earls Court Gym P.O. Box 755 254 Earls Court Road London SW5 9AD Tel: 020-7370 1402

Membership type	Membership Rates		
	Period	Unrestricted access (i.e. anytime during opening hours)	Restricted access (i.e. Mon. to Fri. 9 a.m. to 5 p.m. & anytime at weekends)
Aerobics only <i>Access to all exercise classes and shower, sauna and changing facilities</i>	Membership Charge:	And:	
	12 months	£50	
	6 months	£34	£2.50 per morning or evening class, Mon. to Fri.
	3 months	£24	£3.50* per evening or weekend class
	1 month	£17	
Social membership <i>A special class of membership for the infrequent visitor, giving access to the gym or classes and sauna, etc.</i>	The daily rate for non-members is £3 per morning or afternoon class (Mon. to Fri.) & £5* per evening or weekend class.		
	*An additional charge will be made for classes of more than 1 hour		
	Annual fee of £100		
	and:		
	£3.00 per visit before 5 p.m. on Mon. to Fri. £4.50 per visit after 5 p.m. on Mon. to Fri. and at weekends		
Sunbeds Per session Block of 5 sessions Block of 10 sessions	£6.50		
	£28.50		
	£55.00		
Opening times			
Mon. to Fri. Sat. Sun.	Membership with unrestricted access		Membership with restricted access
	Gym closes		9 a.m. to 5 p.m.
	9 a.m. to 10 p.m. 10 a.m. to 7 p.m. 11 a.m. to 5 p.m.	9.40 p.m. 6.40 p.m. 4.40 p.m.	Anytime
Lockers Lockers are provided in the changing rooms free of charge. A key may be picked up at Reception on production of a current membership card.			

Towels

Towels are available for hire at Reception at 75p each.

Food supplements & sportswear

A range of sportswear and food supplements is available for sale at Reception.

Gym assessments

Bookings may be made at Reception or by ringing 020–7370 1402.

Memberships

Each member will be provided with a membership card which must be produced and surrendered to Reception on each visit.

Memberships are not transferable and no refunds will be made.

Membership (new and renewals) is at the discretion of the Management

Summary

- 1 The relevant costs for decision-making involve opportunity costs, replacement costs and incremental costs.
- 2 The nature of a firm's production function determines the nature of its cost functions, both in the short run and in the long run.
- 3 Managers want to know the type of cost function their firm has in order to make pricing and other marketing mix decisions, and to forecast and plan for the level of costs and inputs for producing a given level of output.
- 4 In the short run the law of diminishing returns causes marginal cost, average variable cost and average total cost to rise beyond a certain level of output.
- 5 The most efficient level of output in the short run for a given plant size involves a trade-off between spreading fixed costs and getting diminishing returns.
- 6 Short-run cost functions can take several mathematical forms: linear, quadratic and cubic.
- 7 Managers want to know the nature of their firm's long-run cost function in order to determine the appropriate plant size.
- 8 In the long run, unit costs are affected by economies and diseconomies of scale, which often cause the long-run average cost curve to be U-shaped.
- 9 Economies of scale are aspects of increasing scale that lead to falling long-run average costs; diseconomies of scale are the opposite. They can thus be regarded as the cost advantages and cost disadvantages for a firm or plant in growing larger.
- 10 Economies and diseconomies can arise at the level of the product, the plant and the firm as a whole.
- 11 Because of uncertainties regarding demand levels there may be a mismatch between the level of output produced by the firm and its sales; there are a number of different strategies that firms can use in this situation.
- 12 Economies of scope arise when there are cost complementarities of producing products together.
- 13 The learning curve describes the situation where unit costs are reduced as cumulative output increases, because of learning better ways of performing a task or tasks.
- 14 CVP analysis analyses relationships between output levels and costs, revenues and profits. It is normally based on the assumption that both total cost and total revenue functions are linear.

Review questions

- 1 Explain the difference between the explicit cost of buying a textbook on economics and the opportunity cost, stating any assumptions. How are these costs relevant for the decision to buy the book?

- 2 Explain the relationship between a firm's short-run production function and its short-run cost function.
- 3 Explain whether the following statements are true or false:
 - a. In the long run a firm might choose to operate a larger plant at less than maximum efficiency rather than a smaller plant at maximum efficiency.
 - b. Maximum efficiency is achieved when *AVC* and *MC* are equal.
 - c. An improvement in technology will shift the *LAC* curve upwards.
 - d. If *AVC* and price stay the same when there is an increase in fixed costs the *BEO* will decrease.
 - e. A firm can have diminishing returns and increasing returns at the same time but not economies and diseconomies of scale.
- 4 Explain how the linearity assumption in CVP analysis compares with conventional economic theory.
- 5 Explain how a firm might gain economies of scope without gaining economies of scale.
- 6 Explain why it is important to distinguish between economies of scale and the learning curve if they both cause unit costs to decrease.

Problems

- 6.1 Quikpak sells returnable containers to major food processors. The price received for the containers is £2 per unit. Of this amount £1.25 is profit contribution. Quikpak is considering an attempt to differentiate its product through quality improvement at a cost of 5p. per unit. Current profits are £40,000 on sales of 100,000 units.
 - a. Assuming that average variable costs are constant at all output levels, find Quikpak's total cost function before the proposed change.
 - b. Calculate the total cost function if the quality improvement is implemented.
 - c. Calculate Quikpak's break-even output before and after the change, assuming it cannot increase its price.
 - d. Calculate the increase in sales that would be necessary with the quality improvement to increase profits to £45,000.
- 6.2 Two business students are considering opening a business selling hamburgers next summer. The students view this as an alternative to taking summer employment with a local firm where they would each earn £3,000 during the three-month summer period. It would cost £2,000 to obtain a licence to operate their stand, £1,000 per month to rent the stand with the necessary equipment and £100 per month for insurance. Petrol costs are estimated at £10 per day. Hamburger meat can be bought for £4.00 per kilo and buns cost £1.20 per dozen. The burgers would sell in 125 gram patties for £1.50 each.

- a. Find the accounting cost function for the proposed business.
 - b. Find the economic cost function for the proposed business.
 - c. Calculate the level of output where the business would make normal profit.
 - d. Cheese slices for the burgers can be bought for £2.40 for twenty slices and it is estimated that 30 per cent of customers would ask for the cheeseburgers, with these selling for £1.95. Calculate the effect of this on the output necessary to make normal profit.
 - e. If the students can sell 150 burgers a day, 30 per cent with cheese, estimate the economic profit they would make. Advise them whether they should enter the business.
- 6.3 Blunt Corporation manufactures calculators, involving a product line of two models. The current price charged for its B1 model is £8, and profit contribution is 25 per cent. Sales have been disappointing for this model, so Blunt is considering a quality improvement at a cost of 50p. per calculator. Advertising would be increased by £40,000 to promote the improvement. Current profit is £60,000 on sales of 80,000 items per month. There is a relationship between sales of B1 and B2 models, such that for every four extra sales of a B1 model there is one less sale of a B2 model, as some consumers switch from one model to the other. The B2 models sell for £6 and have an average variable cost of £5.
- a. Determine the cost function for the B1 model with the changes that Blunt is considering.
 - b. Calculate the increase in sales of B1 models necessary with the quality improvement in order to earn increased profits of £40,000, ignoring the effect on sales of B2 models.
 - c. Calculate the increase in sales of B1 models necessary with the quality improvement in order to earn increased profits of £40,000, taking into account the effect on sales of B2 models.
- 6.4 Haedoo is presently struggling to survive in the motor car industry. Competition is increasing and the company is plagued by overcapacity. Its capacity is 2 million units per year, but it is currently operating at only 70 per cent of this level, and this is resulting in an annual loss of \$480 million. Its profit contribution per unit is 25 per cent. Haedoo has now set targets for its performance in two years' time: it aims to reduce its capacity to 1.5 million units and operate at 80 per cent of this level; it aims to have a profit of \$800 million; and in order to achieve its target output and sell it, it aims to keep its prices the same, while reducing its level of unit variable costs to 90 per cent of their current level by rationalizing its supply procedures and standardizing components.
- a. Calculate the target average level of price for Haedoo.
 - b. Calculate the target level of unit variable costs.
 - c. Calculate the target level of break-even output.

- 6.5 Two firms, PRO and CON, produce a homogeneous product, which they sell at the going market price. PRO has fixed costs that are 40 per cent more than CON and average variable costs of \$21, which is 20 per cent lower than CON's. PRO has a profit contribution of 30 per cent of revenue and a break-even output 4,000 units less than CON's; it produces 10 per cent more revenue than CON and makes 660 per cent more profit than CON.
- Calculate the market price.
 - Calculate the cost functions of both firms.
 - Calculate the outputs of both firms.
 - Calculate the profits of both firms.
- 6.6 Fyrwood is a firm of mortgage brokers which operates an office in London. An analysis of its monthly operating costs indicates that it has average variable costs given by:

$$AVC = 952 - 0.048Q$$

where **AVC** is in £ and **Q** is the number of mortgages sold. Fixed costs are estimated at £36,000 per month. Fyrwood charges a fee of £800 for each mortgage arranged, and 20 per cent of this is paid to its sales consultants.

- Calculate the volume of mortgages that have to be sold in order to make a profit of £60,000 per month.
- Calculate the degree of operating leverage at the above output and interpret your result.

Notes

- See, for example, J. R. McGuigan, R. C. Moyer and F. H. deB. Harris, *Managerial Economics: Applications, Strategy, and Tactics*, 8th edn, Cincinnati: South-Western College, 1999, p. 348.
- F. M Scherer, and D. Ross, *Industrial Market Structure and Economic Performance*, 3rd edn, Boston: Houghton Mifflin, 1990, chap. 4.
- 'Large industrial stoppages in Great Britain', *Employment Gazette*, 86 (January 1980).
- See, 'for example', B. D. Henderson, 'The application and misapplication of the experience curve', *Journal of Business Strategy*, 4 (Winter 1984): 3–9.
- 'Green fuel runs out of gas', *The Sunday Times*, 28 March 2004.
- 'On the rocky road to marriage', *The Economist*, 20 May 2000.